

Evaluating Presence in VR with Self-Representing Auditory-Vibrotactile Input

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ABSTRACT

In this paper, we present the results of an experiment testing various pairings of auditory feedback devices on immersion and emotion in Virtual Reality (VR). We investigate the effects of bone conduction headphones, a chest-mounted vibration speaker, headphones, and combinations thereof, in combination with internal (self-representing) sounds and vibrations in two simulated scenarios. Results suggest that certain auditory-vibrotactile inputs can influence immersion in an intense virtual scene and evoke emotions in a relaxing virtual scene. In addition, self-representing sounds were observed to significantly weaken immersion in the relaxing scene.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques—Auditory feedback;

1 INTRODUCTION AND PRIOR WORK

Up to now, studies that have aimed to improve immersion in virtual environments have mainly focused on visual and tactile input. In contrast, using audio to influence presence in virtual environments has received much less attention by researchers, despite having demonstrable effects on immersion [2]. In particular, sounds that are produced by one's body or avatar (heartbeat, breath, etc.) can have an impact on perception but have been largely overlooked. We call this type of sound a "self-representing" sound, which describes a category of ecological sounds that come from a person's own body and activities. A good example is a heartbeat, which can influence a user's subjective experience of VR, as shown by Chen et al. [1]. Another example from the literature by Nordahl et al. showed that footsteps can have an effect on the perception of motion in VR [3].

Many researchers have already explored techniques for evoking human emotions in virtual environments, especially fear. For example, Robillard et al. [4] evoked anxiety in therapeutic virtual environments derived from computer games. Based on this work, Shibani et al. [5] observed fear renewal with multiple context exposure in virtual environments. Similar to our study, work by Chen et al. [1] tried to display users' real time heart rate by visual/audio/haptic with different emotional virtual experiences, including: happiness, anxiety, fear, disgust, and sadness. They concluded that audio and haptic feedback are more suitable and natural modalities for heart rate representation in VR.

In this study, our goal was to test whether certain types of sounds delivered through various auditory-vibrotactile feedback devices (Fig. 1) can affect factors such as immersion, presence, and emotional state in virtual environments. The devices we evaluated included a pair of bone conduction headphones (bonephones), common headphones (headphones), a chest-mounted vibration speaker, and auditory-vibrotactile combinations thereof. Our hypothesis was that some of the sounds and combinations of hardware would result in an increase in presence or emotional response (fear, relaxation, pleasure, arousal, and/or dominance). In particular, we investigated

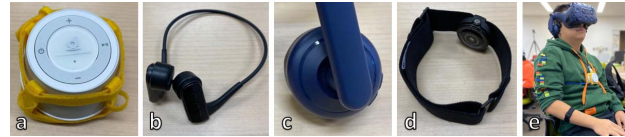


Figure 1: Images of the devices and experiment setup, including a) the vibration speaker, b) bone conduction headphones, c) a speaker from the HTC Vive Pro, d) the polar heart rate monitor, and e) a participant wearing a combination of the auditory-vibrotactile devices.

whether the auditory-vibrotactile self-representing sounds provided by these device sets could enhance immersion and the feeling of presence while immersed in intense and relaxing (Fig. 2) virtual environments. Environmental sounds were also included to investigate the performance of input methods in conjunction with different types of sound. In addition, we observed if there was a significant difference between the bonephones and headphones on immersion and emotion, independent from other factors.

2 EXPERIMENTS

In this experiment, we compared six different input methods to investigate which combinations of auditory and vibrotactile mechanisms would be more effective at enhancing immersion for a given self-representing sound. Additionally, we explored which kinds of sounds might better influence emotion for each input. Twenty-two participants (9 female, 13 male, mean age 25.3, stdev 6.1) took part. All participants had normal hearing and tactile sensing. They were informed that they would experience vibrations on their chests and that heart rate would be collected in the experiment. Furthermore, they were asked if they had heart conditions or any in-body devices. This study was approved by a university ethics committee.

2.1 Stimuli and Design

The intense and relaxing scenes included a dark, abandoned hospital and a tropical beach, which were imported into Unity3D, version 2019.2.8f1, and modified for our experiment. Stimuli were chosen with each virtual environment: Under the intense environment, we chose a relatively fast heartbeat (120 bpm) and evil laughing sound effects as the two stimuli. In the relaxing environment, we selected sleeping (soft breathing with light snoring) and ocean wave sound effects. Audio was presented with all six input methods: vibration (V), headphones (H), bonephones (B), vibration + headphones (V+H), vibration + bonephones (V+B) and headphones + bonephones (H+B). The volume of audio and vibrations were adjusted to the same levels with the windows volume mixer.

Auditory and/or vibrotactile stimuli lasted 1 minute for each scene. The trials for the intense and relaxing scenes were alternated, with each intense trial being followed with a relaxing trial. The relaxing scene was designed to be delivered after each intense scene so that subsequent changes between intensity and relaxation would be larger. The presentation method for each pair of intense and relaxing scenes was the same. The order of these pairs, i.e. the presentation methods, was randomized. In total, 12 pairs (6 methods times 2 kinds of audio for both scenes), a total of 24 trials, were tested for this experiment. After the presentation of each method,

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Figure 2: Images of the VR environments, showing the abandoned hospital that was used as the intense scene and tropical beach that was used as the relaxing scene in experiments.

participants conducted a survey including subjective questions and a self assessment mannequin (SAM).

3 RESULTS

Heart rate data was analyzed with ANOVA. SAM ratings (self-reported pleasure, arousal and dominance) and subjective ratings of immersion, fear, and relaxation were analyzed with a non-parametric Friedman test and Penalized quasi-likelihood using the Generalized Linear Mixed Model (GLMMPQL), written in R.

Most of the heart rate data we collected was visibly stable over time, and changes in participants' heart rates were relatively small. ANOVA revealed no significant effect of environment or sound on participant heart rate.

3.1 Ratings of Immersion, Fear, and Relaxation

Results from the GLMMPQL analysis of immersion ratings of the intense scene suggest that the method of sound delivery had an effect on participants' feelings of immersion. The V+B (estimate = 0.086, $p < 0.05$) and V+H (estimate = 0.102, $p < 0.01$) were higher rated for immersion. There was no significant difference between the two sound types in this condition. No significant interactions were revealed between sound delivery methods and feeling of fear.

Participants' feelings of immersion were affected by sound types: the self-representing sound (estimate = -0.153, $p < 0.0001$) was less effective at enhancing immersion than the environmental sound.

As the result of applying GLMMPQL to subjective ratings of relaxation under the relaxing scene, a significant effect on feeling of relaxation was found between sound types. The self-representing sound (estimate = -0.214, $p < 0.0001$) was less effective on enhancing immersion than environmental sound. No significant interaction was found between sound delivery methods. Subjective feelings of immersion and fear were not found to be significant.

3.2 Effects on Self-reported Emotional Experience

All subjective and SAM ratings were analyzed with non-parametric Friedman tests and Wilcoxon signed-rank tests with Benjamini-Hochberg correction. In the **intense scene** with environmental sound (Fig. 3), a significant effect was found for pleasure rating ($\chi^2 = 19.227$, $df = 5$, $p < 0.01$). These showed significant differences for V vs. B ($V = 21$, $p < 0.05$), H vs. V+B ($V = 62$, $p < 0.05$) and B vs. V+H ($V = 148.5$, $p < 0.05$). With self-representing sound stimulus (Fig. 3), a significant effect was found for arousal rating ($\chi^2 = 18.313$, $df = 5$, $p < 0.01$) and for H vs. V+B ($V = 0.001539$, $p < 0.05$). In the **relaxing scene** with the self-representing sound (Fig. 4), a significant effect was found for pleasure rating ($\chi^2 = 16.438$, $df = 5$, $p < 0.01$) for V+B vs. H+B ($V = 153$, $p < 0.05$).

With environmental sound stimuli (Fig. 4), a significant effect was found for pleasure rating ($\chi^2 = 25.736$, $df = 5$, $p < 0.001$) and for V vs. H ($V = 13$, $p < 0.01$), V vs. B ($V = 4.5$, $p < 0.01$), V vs. V+B ($V = 22$, $p < 0.05$), V vs. H+B ($V = 10$, $p < 0.01$), V+H vs. H+B ($V = 4$, $p < 0.05$), V+B vs. H+B ($V = 10$, $p < 0.05$). For self-reported feelings of dominance in the SAM ratings no significant effects were found in intense or relaxing scenes. We applied GLMMPQL to the self-reported SAM ratings for the intense scene. Results suggested that method of sound delivery had an effect on pleasure. This feeling was weakened with V (estimate = -0.162, $p < 0.01$) and V+H (estimate = -0.109, $p < 0.05$). For arousal, H

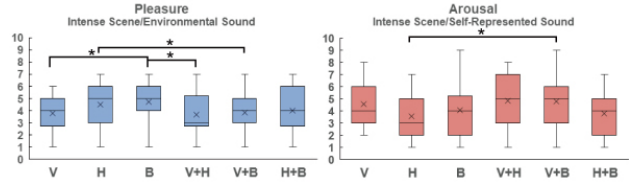


Figure 3: Self reported pleasure with the environmental sound and arousal with the self-representing sound in the intense scene.

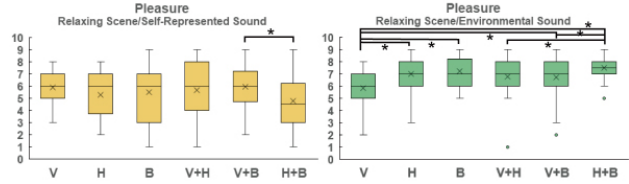


Figure 4: Self reported pleasure with the self-representing sound and pleasure with the environmental sound in the relaxing scene.

(estimate = -0.168, $p < 0.05$) had less of an effect on the feeling. In addition, self-representing sounds (estimate = 0.094, $p < 0.05$) were more effective compared to environmental sounds. For the relaxing scene, V (estimate = -0.1, $p < 0.05$) was less effective, while the self-representing sound (estimate = -0.203, $p < 0.0001$) was less effective than the environmental sound. No significant interaction was found between methods and sound types.

4 CONCLUSION

Based on our experimental results, certain self-representing sounds and auditory-vibrotactile inputs can potentially enhance immersion and evoke emotions in VR. In addition, the actual tasks and interactions that participants complete in an experiment may have a cumulative effect when combined with various sounds, further affecting participants' feelings of presence. As such, self-representing sounds deserve further consideration as a tool to enhance immersion, especially in studies involving a self-avatar.

As intense virtual experiences with auditory-vibrotactile inputs tend to be more emotionally evoking and engaging, training simulators for athletes, performers and rescuers could benefit from enhanced sound to provide more realistic feelings of stress or nervousness. On the other hand, since self-representing sounds could also weaken immersion during relaxation, these inputs might be helpful for improving situational awareness.

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